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Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements

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ABSTRACT

This paper investigates the vortex-induced vibration of a twin steel box girder suspension bridge with a centre span of 1650 m based on field measurements. Two ultrasonic anemometers, two tri-axial accelerometers and 52 wind pressure sensors are installed at the quarter span section. The other 20 pressure sensors are installed in another 5 sections, and each section has 4 pressure sensors. Four vortex-induced oscillation events are measured. The analytical results indicate that the vortex-induced vibration more likely occurs in a low wind speed range of 6–10 m/s, with the wind direction nearly perpendicular to the bridge line, and low turbulence intensity. The mean pressure distribution on the surface of the bridge deck is obtained and the characteristics of fluctuant pressures are analysed by proper orthogonal decomposition (POD) method. Moreover, the spatial–temporal evolution of flow around the bridge deck is investigated. The results indicate that in the beginning stage of vortex-induced resonance, the regular vortex shedding phenomena occur only in the gap of the deck and at tailing region of downstream deck; however, when in the lock-in stage, the vortex shedding is strengthened due to the dramatic vibration, and the regular vortex shedding phenomena extend to the entire lower surface of downstream deck and the tail of upstream deck, the vortex shedding regions in the gap and lower surface link together. In the lock-in range, the span-wise correlation of the wind pressure is analysed, and the correlations of wind pressure along the bridge line are very high and do not decrease with the increase in distance.

1. Introduction

With increases in the spans of bridges, bridges become more flexible and small damping occurs. Therefore, the bridges more readily oscillate dramatically when subject to wind. The vortex-induced vibration (VIV) is just a typical phenomenon of wind-induced vibration, especially for the long-span bridges. When flow passes through the bridge deck, vortex shedding occurs. If the vortex shedding frequency is close to the natural frequency of the bridge, it can cause vortex-induced resonance. Although vortex-induced vibration is limited amplitude vibration and does not cause collapse, it can result in large displacements and discomfort to the drivers. In addition, the vortex-induced oscillations commonly occur in the low wind speed region, so the occurrence probability of vortex-induced vibration is high, resulting in long-term fatigue damage. In the previous studies, wind tunnel test is employed to investigate vortex-induced vibration. However, this technology has limitations, e.g., due to the size effect of the wind tunnel, it cannot simulate the vortex-induced vibration for a high Reynolds number; the damping of the real bridge is also difficult to simulate in the wind tunnel test. Field measurement just can overcome the limitations of the wind tunnel test, although it has some shortages, such as cost and uncontrollable wind conditions. Field monitoring provides excellent opportunities to investigate the full-scale vortex-induced vibration of bridges and the mechanism of structure and flow interaction. On the other hand, it also can validate the wind tunnel test results.

In previous research, the field monitoring for wind and wind effects on tall buildings has been comprehensively conducted [1–6]. However, the investigation of wind effects on long-span bridges based on field measurements is very rare, especially for the full-scale vortex-induced vibration. Frandsen [7] investigated the vortex-induced oscillations of the Great Belt East Bridge (one single steel box girder section) based on full-scale measurements. In that article, the wind pressure at the deck surface and accelerations were simultaneously measured. The phenomenon of cross-wind vortex-induced oscillation was observed in smooth flow, a low wind speed (around 8 m/s) and a corresponding direction perpendicular to the bridge line. The estimated full-scale

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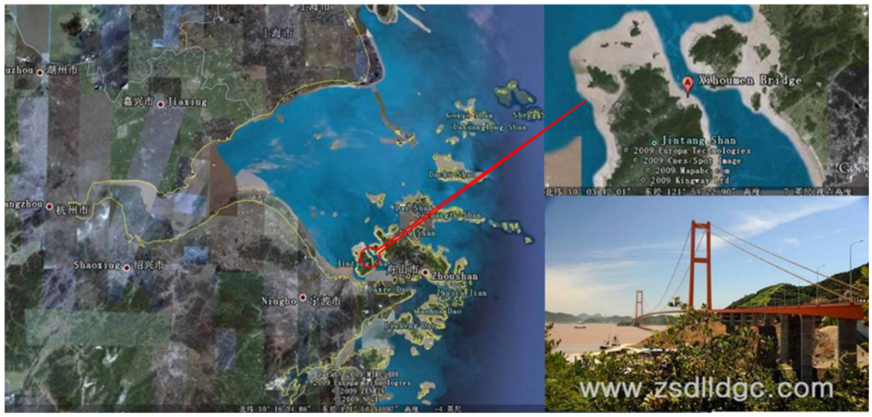

Fig. 1. Location of the investigated bridge.

Strouhal number of the deck lies in the range 0.08–0.15. According to the pressure sensors on the deck surface, the time-averaged pressure coefficients were near zero on the top flange trailing edge. Fujino and Yoshida [8] investigated the vortex-induced vibration in ten-span continuous one single steel box girder bridge section of the Tans-Tokyo Bay Crossing Bridge. The first-mode vertical vortex-induced vibration was observed with the wind directions within $\pm 20^{\circ}$ of the bridge transverse axis. When the wind velocity reached around 16–17 m/s, the maximum amplitude of the first-mode vertical vortex-induced vibration exceeded 50 cm. The wind tunnel tests on two-dimensional sectional models and three-dimensional bridge models were also conducted. The results of the field and wind tunnel tests were consistent with each other. To control the first and second vertical vibrations of the bridge, a new type of TMD was developed and exhibited sufficient vibration control.

With the increase in span length, twin-separated steel box girder section will be extensively used in long-span bridges to improve bridge aerodynamic stability. However, the investigations on the wind-induced vibration of this kind of bridges are insufficient so far. The investigated bridge is a suspension bridge with a main span of 1650 m. A two separated steel box girder section is employed in this bridge because both typhoon in summer and seasonal wind in winter are very strong in the site of the bridge. A sophisticated structural health monitoring (SHM) system is designed and installed on this bridge, and sensors for monitoring of wind, wind pressure and wind-induced vibration are included in the SHM system. The SHM system provides a tool to investigate the wind-induced oscillation of the prototype long-span bridge with twin-separated steel box girder and the results can be further compared with those from the wind tunnel test and the computational fluid dynamics (CFD) numerical simulation, which will aid to understanding the aerodynamic effects of this kind of bridge under different scales. Since the bridge was completed, vortex-induced vibration of this bridge has been observed and recorded by the SHM system. This paper presents the behaviour of the vortex-induced oscillation of the bridge with two separated steel box girder based on field measurement. The main objective of this study is to investigate the full-scale vortex-induced vibration of the bridge based on field measurements.

2. Description of the bridge and field measurement system

2.1. Description of the bridge

The investigated bridge is a cross-sea bridge in the East China Sea. The location of this bridge is shown in Fig. 1.

The Bridge is an asymmetric suspension bridge with a 1650 m central main span (ranking No. 2 in suspension bridges around the world) and one 578 m side span. The north and south towers are made of reinforced concrete, each with a height of 236.5 m. The bridge deck is composed of two separated steel box girders with a deck centre depth of 3.51 m. The gap width between two box girders is 6 m, and the total width of the bridge deck is 36 m. The configuration of the deck is presented in Fig. 2.

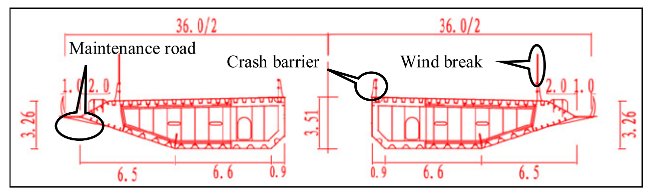

Fig. 2. Cross section of the separated box girder (unit: m).

2.2. Field measurement system

Typhoon and seasonal winds are very strong near this bridge, so a wind monitoring system for this bridge is set up. Two three-dimensional ultrasonic anemometers (Young Model 8100), as shown in Fig. 3(a), are installed on the east and west lighting columns at the quarter span section of the deck, and each anemometer is 6 m above the deck surface and 54.5 m above the sea surface. This type of anemometer has a good resolution (wind speed: 0.01 m/s, wind direction: 0.1°) and high accuracy (wind speed: $\pm 1\%$, wind direction: $\pm 2^{\circ}$), and the measurement wind speed range is (0–40 m/s). Two force-balance tri-axial accelerometers (GT02, the measured range is ± 2.0 g, the sensitivity is 2.7 V/g, and the frequency band is DC $\sim$ 120 Hz), as shown in Fig. 3(b), are installed on the east and west sides of the bridge deck to measure the lateral, vertical and torsion accelerations of the bridge deck; the horizontal distance between them is 30 m. To obtain the wind pressure on the deck surface, 72 wind pressure sensors (CY200FA1P), as shown in Fig. 3(c), are implemented at the bridge surface at different sections, as shown in Fig. 4(a) and (b). At quarter span section S1, 52 wind pressure sensors are installed around the section, and the detailed locations of the sensors are shown in Fig. 4(c). The other 20 pressure sensors are installed at another 5 sections (S2, S3, S4, S5, S6) to investigate the span-wise correlation of the wind pressure field; each section has 4 pressure sensors, two are installed on the east side, and the other two are installed on the west side. The locations of these 5 sections are shown in Fig. 4(b). The type of wind pressure sensor is a kind of diffused silicon pressure sensor with a measured range of ± 1.5 kPa, a resolution of 1 Pa, and an accuracy of $\pm 0.25\%$. The frequency band is 0–1 Hz. The sensor size is $10 \times 5 \times 4$ cm, and diameter of the gas nozzle is 8 mm. To investigate the influences

Fig. 3. Sensors used in the wind and wind effect monitoring system: (a) Young Model 8100 ultrasonic anemometer, (b) tri-axial accelerometer and (c) wind pressure sensor.

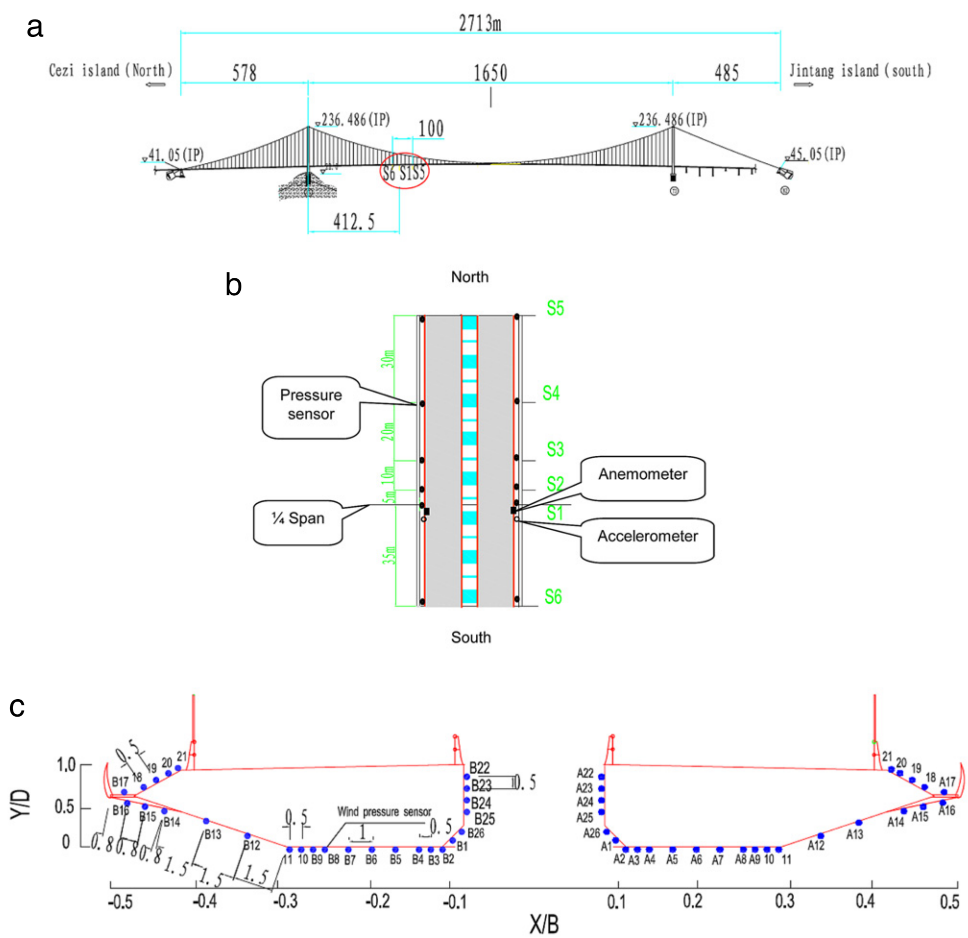

Fig. 4. Layout of the sensors: (a) Elevation drawing, (b) Top view and (c) Locations of pressure sensors around the bridge deck at section S1.

of the sensor on the flow field, a calibration test is conducted in the wind tunnel of Harbin Institute of Technology. The wind pressure of free flow is measured by the pressure sensor and pressure scanner (DSM3400) at the same conditions and the results are shown in Fig. 5. It can be seen from Fig. 5 that the pressure measured by these two kinds of sensors agreed well with each other. Therefore, the disturbance of the pressure sensor on the flow field can be ignored.

The field measurements were conducted from September 29th, 2009 to November 30th, 2009. During the measured period, the bridge did not open to public traffic (just in the interval of bridge completion and before opening to public traffic). Therefore, the influence of traffic on the bridge dynamic behaviours is free. The sampling frequencies are set at 4 and 20 Hz for the ultrasonic anemometers and accelerometers, respectively. For the wind pressure sensors, the sampling frequency is set at 10 Hz.

3. Wind climate at the bridge site and acceleration response of the bridge deck

3.1. Wind climate and acceleration response

The ranges of the 10 min mean wind speeds and directions from September 29th, 2009 to November 30th, 2009 are presented in Fig. 6. During this time, the most frequent winds came from the northwest and usually had high wind speeds. Sometimes the winds also came from the southeast, with the wind direction nearly perpendicular to the longitudinal bridge axis. Fig. 7 shows the relationship between the vertical acceleration of the deck and the 10 min mean wind speed. The results indicate that the vertical acceleration of the deck increases with the increase in the mean wind speed. However, it is worth noting that the vertical acceleration of the deck increases significantly in the wind speed

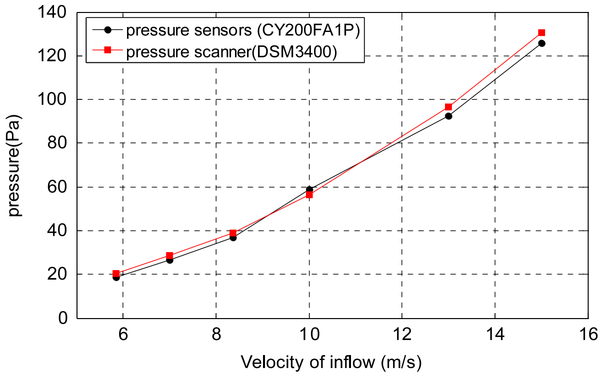

Fig. 5. The pressure measured by pressure sensor (CY200FA1P) and pressure scanner (DSM3400).

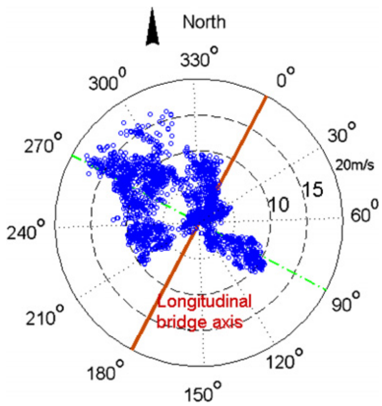

Fig. 6. Wind rose diagram during the measured period (20/9/2009–30/11/2009).

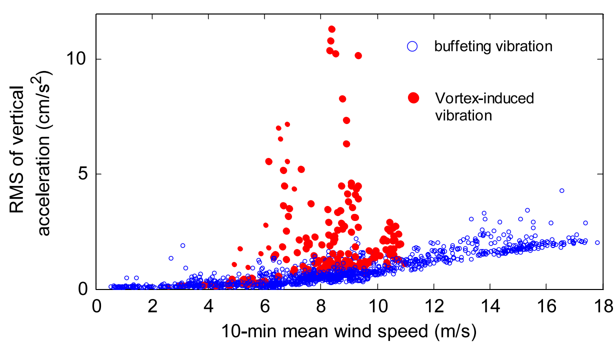

Fig. 7. Relationship between vertical acceleration of the deck and 10 min mean wind speed.

range of 6–10 m/s, i.e., the vortex-induced vibration of the bridge occurs in the wind speed range of 6–10 m/s. The detailed vortex-

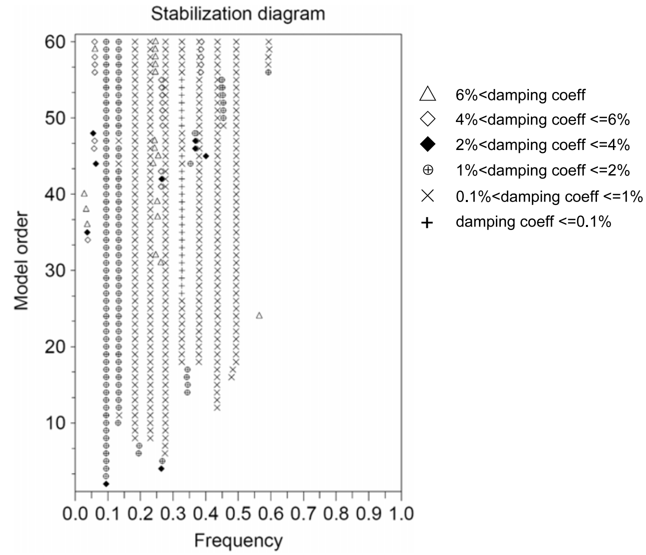

Fig. 9. Stabilization diagram.

induced vibration phenomena will be analysed in the following sections.

3.2. Modal analysis of the bridge

The modal parameters of the bridge are very important for investigating the mechanism of the wind-induced vibration of a bridge. The stochastic subspace identification (SSI) method is a powerful tool to identify the modal parameters for the output-only system under white noise excitation [9–11]. Because the wind load is a wide frequency band stochastic process and can be regarded as a white noise process, the data-driven SSI is employed to identify the modal parameters of the bridge in this study. The measured acceleration data is excluded when vortex-induced vibration occurs in this bridge, and the data is divided into hourly segments. A representative segment of measured vertical acceleration of bridge deck is shown in Fig. 8. The range of analysed frequency is 0.01–0.6 Hz because high mode oscillation is not observed. The dimension of block Hankel matrix is 500 × 500. Fig. 9 shows the stabilization diagram obtained from a 1 h measured acceleration (as shown in Fig. 8) when the wind speed is $U = 7.92$ m/s with a direction of $\theta = 302.4°$ and turbulence of $I_u = 14.95\%$. The natural frequencies and damping ratio can be obtained from Fig. 9. As is well known, the identified modal parameters based on field measurement consist of structural dynamic parameters and aerodynamic parameters. Fig. 10 shows the relationship between the measured first five vertical damping ratios and 1 h mean wind speeds. It can be seen that, for $U > 6.0$ m/s, except the 2nd vertical modal damping ratio, the total damping ratios have an observable increasing tendency with the increase in hourly mean wind speed, implying that the aerodynamic damping increase are considerable (the vortex-induced vibration is not included in this

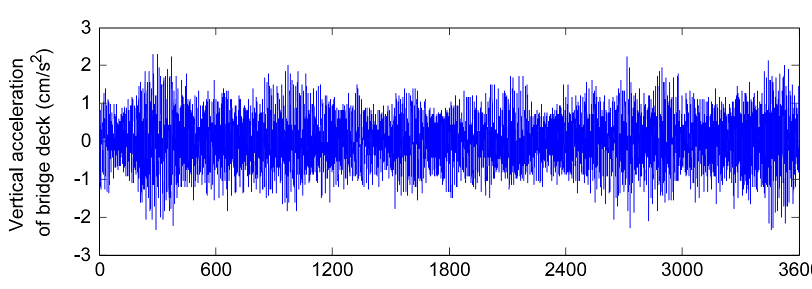

Fig. 8. Time history of a 1 h vertical acceleration of bridge deck.

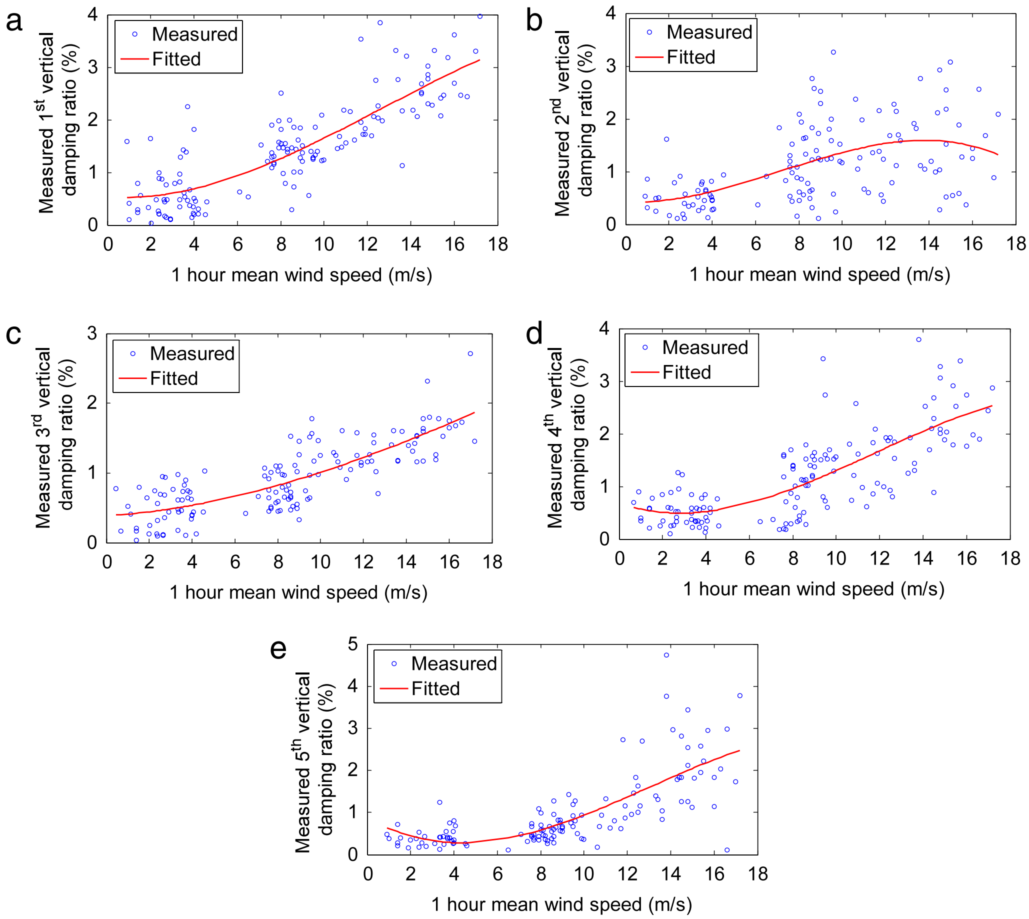

Fig. 10. Relationship between the identified first 5 vertical modal damping ratios and 1 h mean wind speed: (a) The 1st vertical modal damping ratio, (b) The 2nd vertical modal damping ratio, (c) The 3rd vertical modal damping ratio, (d) The 4th vertical modal damping ratio and (e) The 5th vertical modal damping ratio.

Table 1
The measured natural frequencies and modal damping ratios from vertical vibration of the deck.

Order of modes	$U \le 4.0$ m/s	
	Natural frequency (Hz)	Damping ratio (%)
1	0.0953	0.57
2	0.1328	0.52
3	0.1825	0.50
4	0.2301	0.51
5	0.2757	0.39
6	0.3237	0.42
7	0.4351	0.37
8	0.4918	0.32

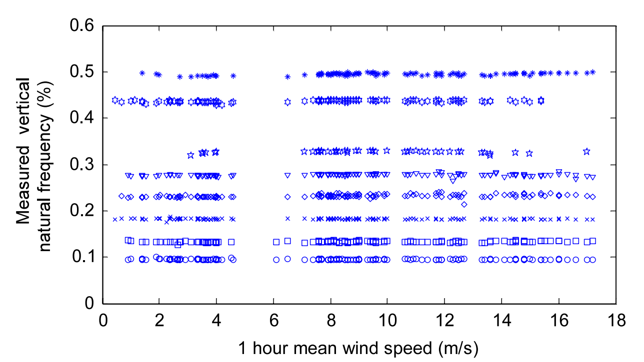

Fig. 11. Relationship between the measured natural frequencies and 1 h mean wind speed.

analysis). However, for $U \le 4.0$ m/s, the wind effects on the damping ratio are very slight, so the identified damping is close to the structural damping ratio. Fig. 11 shows the relationship between the measured natural frequencies and 1 h mean wind speed. It is observed that the natural frequencies of the bridge are independent of the wind speed, i.e., the aerodynamic force has only a slight impact on the structural modal frequencies. Table 1 presents the natural frequencies and modal damping ratios identified from the measured vertical vibration of the deck in the low wind speed range ($U \le 4.0$ m/s), which represent the natural frequencies and damping ratios of the bridge self.

4. Observed vortex-induced vibration phenomena and findings

When the flow passes through the bluff body, vortex shedding may occur. As the flow speed is increased, the vortex shedding frequency approaches the natural frequency of bluff body, dramatic resonance oscillation occurs, in this case, the wind velocity is defined as critical wind velocity; after that, the vortex shedding is in lock-in range and is controlled by large body motion, the vortex shedding frequency is close to the natural frequency of the bluff body, and the bluff body presents limit-cycle oscillation. In the field measurements duration, the vortex-induced vibration occurs on several occasions. In the paper, four vortex-induced vibration events, i.e. events 1#, 2#, 3# and 4#, are analysed. The detailed information about these four VIV events is listed in Table 2 and shown in Fig. 12. The corresponding power spectra indicate that the vertical oscillations are typical harmonic vibrations with a single frequency, and these vibration frequencies are identical to the natural frequencies of the bridge, i.e., 0.1828, 0.2297, and

Table 2
The observed vortex-induced vibration events.

Vortex vibration events	Dominant response frequency (Hz)	Mean wind speed (m/s)	Mean wind direction (°)	Turbulence intensity (%)	Max 10 min R.M.S. of vertical acceleration (cm/s^2)	Duration of VIV (min)
1	0.1828	6.12–7.27	89.0–93.5	2.51–4.01	5.57	30
2	0.1828	6.43–7.01	100.0–103.0	2.84–6.43	7.05	40
3	0.2297	8.31–8.89	94.5–100.0	1.81–5.57	11.33	100
4	0.2767	9.33–8.70	271.1–283.2	4.67–6.64	4.67	60

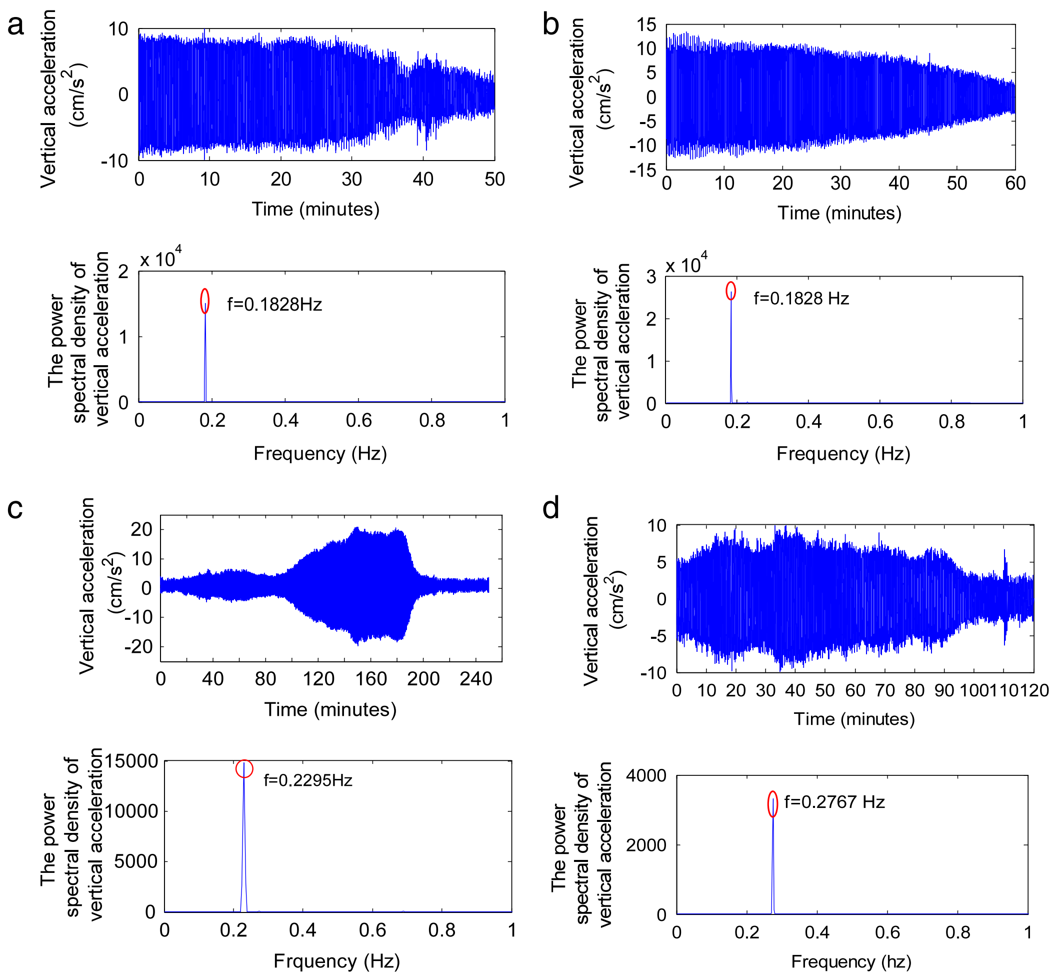

Fig. 12. The vertical acceleration time histories and corresponding spectra: (a) event 1#, 4:00–4:50 pm, September 25th, 2009, (b) event 2#, 6:00–7:00 pm, September 25th, 2009, (c) event 3#, 8:00–11:40 pm, October 29th, 2009, and (d) event 4#, 3:00–5:00 pm, October 2nd, 2009.

0.2767 Hz, corresponding to the measured 3rd, 4th and 5th vertical modal frequencies of the bridge, respectively. Thus, it is reasonable to define these oscillations as vortex-induced oscillations. When vortex-induced vibration is occurring, the wind speeds are over the range of 6–10 m/s, with a corresponding Reynolds number range of $1.44 \times 10^7 \sim 2.40 \times 10^7$ ($\mathrm{Re} = UB/\upsilon$, υ is the kinematic viscosity and assumed constant with a value of 1.5×10^{-5} m^2/s for air at 20 °C, B is the deck width, and $B = 36$ m for this bridge). The wind directions, which vary from 89° to 103° and from 274° to 283°, as shown in Fig. 13, are nearly perpendicular to the bridge line. The corresponding turbulence intensities are all below 7% (turbulence intensity is defined as: $I_i = \sigma_i/U$, $i = u, v, w$, corresponding to longitudinal, lateral and vertical fluctuant wind, σ represents the standard deviation of the fluctuant wind). From the above analysis, the condition in which the vortex-induced vibration of the suspension bridge is more likely to occur can be attributed to a flow with a low wind speed, a low turbulence intensity and a direction nearly perpendicular to the longitudinal bridge axis.

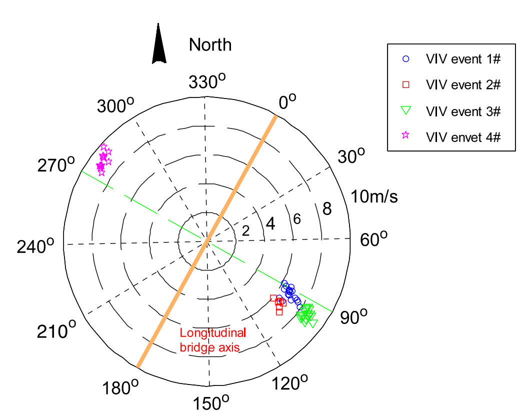

Fig. 13. The 10 min mean wind speeds and directions when vortex-induced vibration occurs.

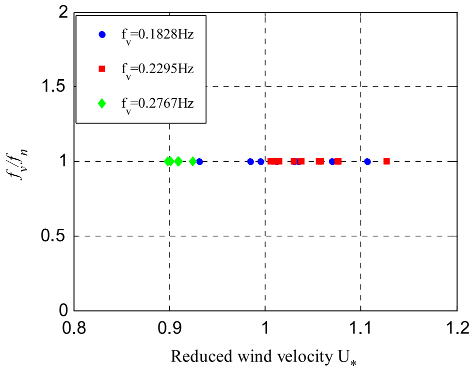

Fig. 14. Correlation of the reduced wind and frequency ratio (f_v is vortex shedding frequency, f_n is the vertical natural frequency of bridge deck).

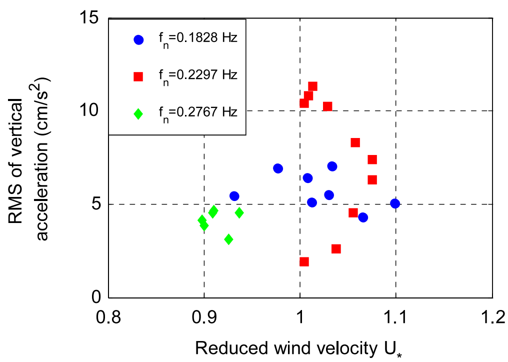

Fig. 15. Relationship of the reduced wind velocity, vortex-induced resonance frequency and RMS of vertical acceleration in the lock-in range.

The correlation of the reduced wind velocity and frequency ratio f_v/f_n is presented in Fig. 14, where the reduced wind velocity is defined as $U_* = U/fB$ (where U is the 10 min mean wind speed, f is the resonant response frequency, B is the deck width and $B = 36$ m). When the vortex-induced resonance frequency $f_n = 0.1828$ Hz, the range of lock-in reduced velocity is $0.93 \leq U_* \leq 1.11$, while for $f_n = 0.2297$ Hz, $1.01 \leq U_* \leq 1.08$ and for $f_n = 0.2767$ Hz, $0.87 \leq U_* \leq 0.94$. Fig. 15 shows the relationship of reduced wind velocity and RMS of vertical acceleration. It is observed that the large acceleration appears when the reduced wind velocity is close to 1.

The Strouhal number St is defined as

$$St = \frac{f_v D}{U}, \tag{1}$$

where f_v is vortex shedding frequency, D is the deck depth, and $D = 3.51$ m in this study, U is the inflow velocity.

The Strouhal number, non-dimensional vortex shedding frequency, is an important factor to understand the aerodynamic character of a bluff body, especially in estimating the critical wind velocity of VIV and the amplitude of the vortex-induced oscillation. As presented in Eq. (1), in order to estimate the Strouhal number St, the wind velocity and the vortex shedding frequency should be measured. The vortex shedding frequency is identified by FFT analysis on the measured pressure. In wind tunnel testing, it is very easy to obtain the wind velocity and the vortex shedding frequency, however, for field measurement, the wind conditions cannot be controllably, and the vortex shedding signal outside the VIV range is very weak, the pressures in wake flow display broadband frequency characteristics. So outside the VIV range, estimating the Strouhal number is very difficult. In VIV range, because of interaction between the vortex shedding and bridge vibration, the vortex shedding signal is strengthened, the pressures in wake flow have

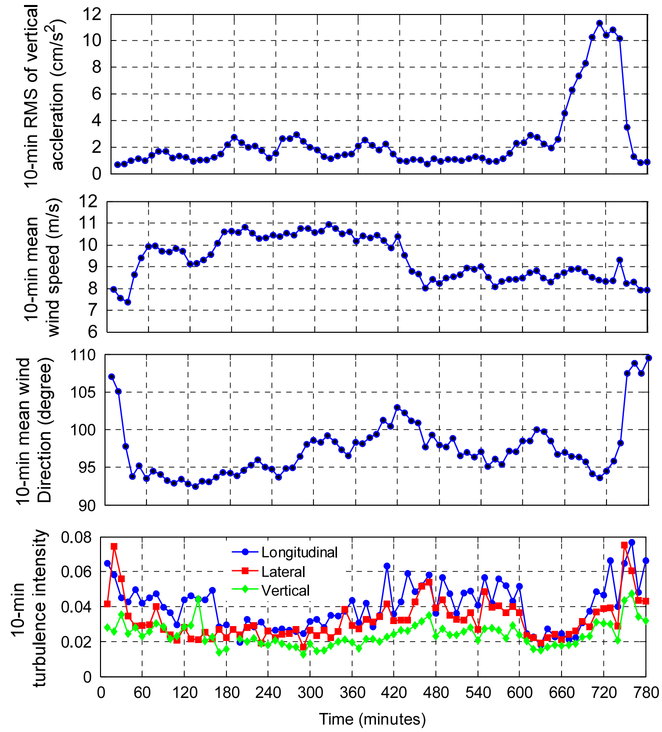

Fig. 16. The 10 min RMS of vertical acceleration of the deck and the corresponding 10 min mean wind speed, direction and turbulence intensity.

obvious dominated frequency. According to the definition of VIV, the Strouhal number can be estimated based on vortex shedding frequency of lock-in range and critical wind velocity, however, for field measurement, the wind speed cannot be controllably gradually increased, the wind speed recorded may be in the lock-in range rather than the critical wind speed. So it is difficult to determine the exact Strouhal number of bridge deck based on the full-scale measurement. Fortunately, for the investigated Bridge, the ranges of the lock-in reduced velocity are very narrow. Therefore, the Strouhal number can be approximately estimated based on the average value of the lock-in range. According to vortex-induced vibration in events 1# and 2#, the approximately estimated $St = 0.0959$; for event 3#, the estimated $St = 0.0942$; and for event 4# $St = 0.1068$. Although some scatter presents, the Strouhal numbers do not change a lot and almost remain a reasonable constant (0.0942–0.1068) within the Reynolds number range of $1.74 \times 10^7 \sim 2.40 \times 10^7$ recorded in the measurement period.

For these four vortex-induced vibration events, the amplitude of vortex-induced vibration for event 3# is largest, so the vortex-induced vibration event 3# only is selected to be analysed due to the space limitation of the paper.

5. Effects of wind conditions on the response behaviour

In this section, to understand the bridge responses in different wind conditions, 13 h (11:00 am–12:00 pm, October 28th, 2009) response data of bridge deck and corresponding wind conditions are first analysed. Fig. 16 presents the 10 min RMS of the vertical acceleration and the corresponding 10 min mean wind speed, mean wind direction and turbulence intensity during this measurement period.

To analyse wind effects on the oscillation process of bridge, the vibration is divided into 5 stages as follows:

Stage 1: the duration is 0–300 min, the wind condition is $U = 9.15$–10.82 m/s, $\theta = 92.5°$–$98.5°$, $I_u = 2.44\%$–4.94%, and the wind speed presents an increasing tendency and then

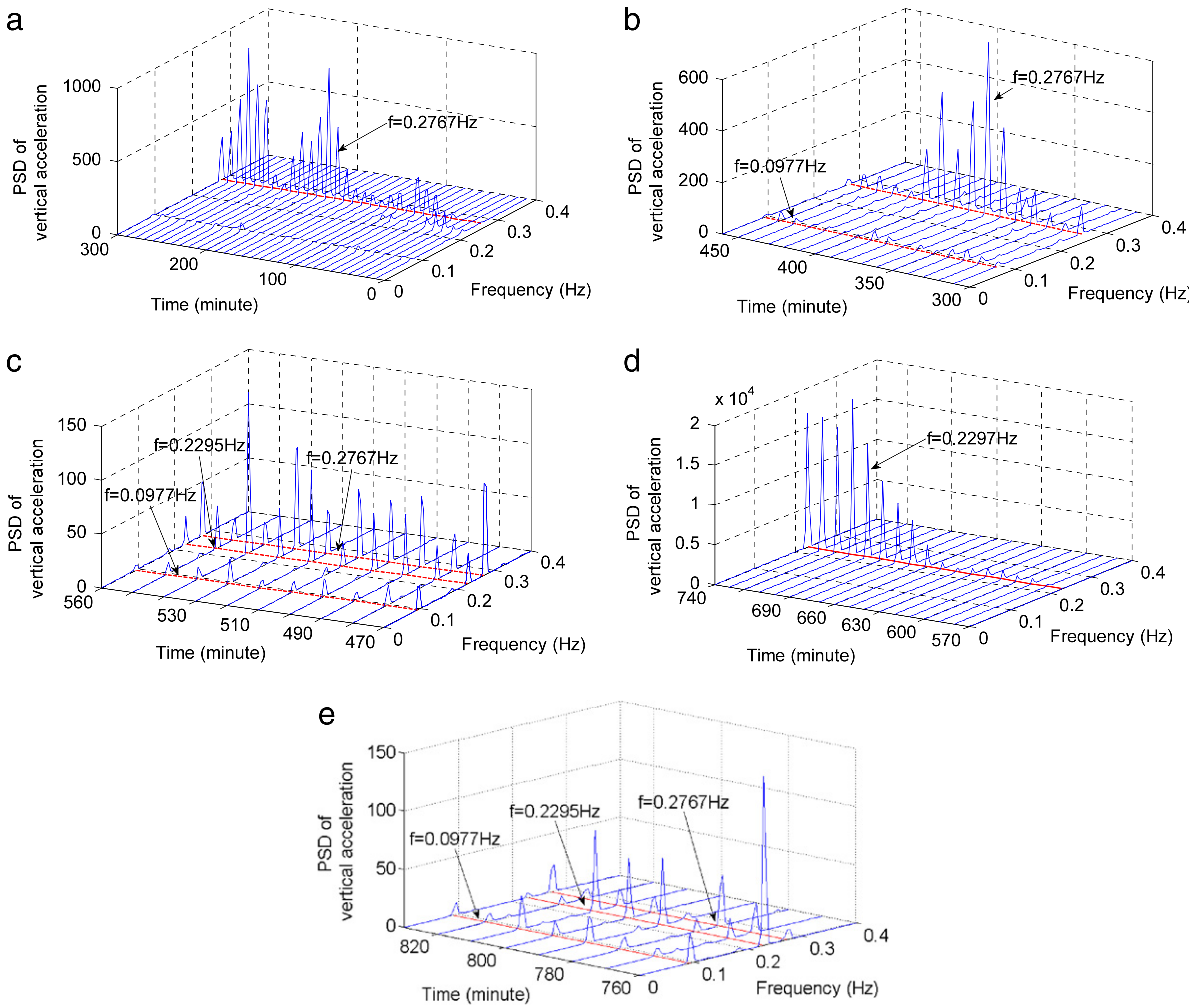

Fig. 17. The power spectrum of vertical acceleration: (a) stage 1, (b) stage 2, (c) stage 3, (d) stage 4, and (e) stage 5.

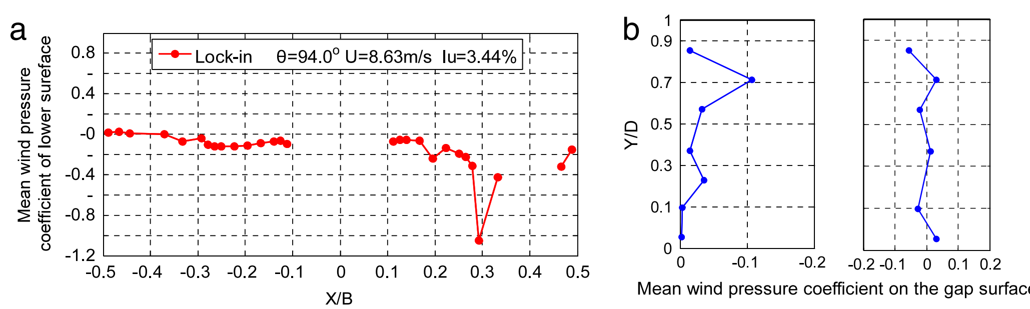

Fig. 18. The 10 min mean pressure distribution in the lock-in range ($U = 8.7$ m/s, resonance frequency = 0.2297 Hz): (a) cross section of the girder and (b) mean pressure distribution.

approximate constant. Under these wind climate conditions, the vertical vibration of the bridge is dominated by the measured 5th mode with the natural frequency of 0.2767 Hz, as shown in Fig. 17(a).

Stage 2: the duration is 300–460 min, $U = 10.63 - 8.02$ m/s, $\theta = 92.5°–98.5°$, $I_u = 3.28\%–6.63\%$, and the wind speed decreases with time. As shown in Fig. 17(b), lower frequencies, e.g., 0.0977 Hz, appear in the vibration, and the contributions of these frequencies to the vibration are considerable in some durations.

Stage 3: the duration is 460–560 min, $U = 8.42–9.01$ m/s, $\theta = 95.1°–99.3°$, and $I_u = 3.60\%–5.82\%$. In this stage, the vibration is dominated by both the measured 4th mode with the natural frequency of 0.2297 Hz and the measured 5th vertical mode with the natural frequency of 0.2767 Hz (see Fig. 17(c)).

Stage 4: the duration is 560–740 min, i.e., the vortex-induced vibration stage, $U = 8.41–8.89$ m/s, $\theta = 94.5°–100.0°$, and $I_u = 1.81\%–4.86\%$., the vibration is dominated by the measured 4th mode with the natural frequency of 0.2297 Hz over entire this stage, as shown in Fig. 17(d), while the amplitude of the acceleration starts to dramatically increase at about 650 min.

Stage 5 (final stage): (See Fig. 17(e).), because the wind direction changes and is not perpendicular to the longitudinal bridge axis, the vortex-induced vibrations vanish.

Comparing the flow characteristics during this whole development process of vortex-induced vibration, the dominant factors on

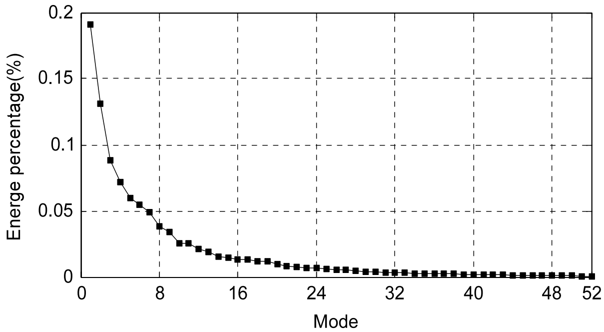

Fig. 19. The energy distribution at each mode.

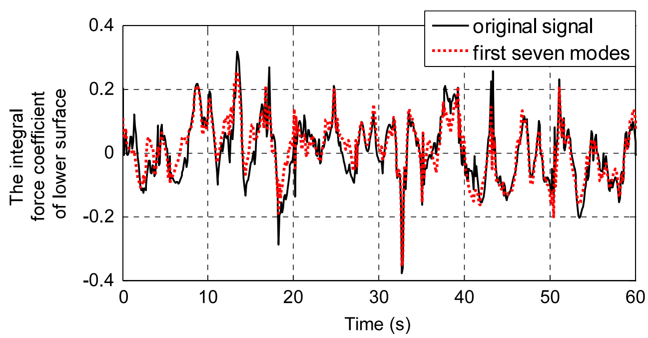

Fig. 20. The time history of the integral force coefficient and reconstructed signal of lower surface.

the vortex-induced vibration may be wind speed, wind direction and wind turbulence intensity. During 600–690 min (occurrence of vortex-induced vibration), the turbulence intensities of longitudinal fluctuant wind maintain small values (from 1.81% to 3%), implying that the turbulence intensity has a significant impact on the occurrence of vortex-induced vibration.

6. Wind pressure distribution in the lock-in region

The wind pressure coefficient is defined as

$$C_p = \frac{p - p_0}{12\rho U_0^2}, \qquad (2)$$

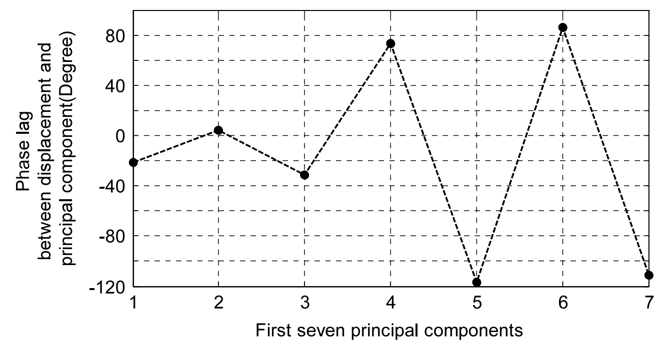

Fig. 22. The phase lag between the displacement and principal component.

where p_0 is the 10 min mean pressure of the reference position in the paper; the reference position is 6-m-high away from the deck surface, U_0 is the 10 min mean wind speed at the reference place, p is the wind pressure at the measured position, ρ is the air mass density, and C_p is the wind pressure coefficient.

Fig. 18 presents a 10 min (10:40–10:50 pm, October 29th, 2009) mean pressure distribution around the bridge deck. The analysed data is from in the lock-in region of VIV in event 3#, and the corresponding wind conditions are: $U = 8.41$ m/s, $\theta = 94.0°$ and $I_u = 4.91\%$. The locations of pressure sensors around the bridge are shown in Fig. 4(c). On the lower surface of upstream deck ($0.1 \le X/B \le 0.5$), all of the pressures are negative and on the corner (O), the pressure coefficient reaches the maximum negative value. In the following range of $0.33 \le X/B \le 0.25$, the pressures present an increasing tendency, the pressure recovers rapidly. The mean pressure coefficients are close to zero at the lower surface of downstream deck, especially at the trailing edge of the deck. On the upstream wall of the gap, the pressure coefficients are close to zero, and on the downstream wall of the gap, the pressure coefficients are negative and their absolute values are very small.

The published studies on the aerodynamics of separated box girders (including wind tunnel test, CFD simulation, and full-scale field) are very scarce. Therefore, further comparison of the wind tunnel test results with the field monitoring results on the two separated box girders is not conducted in the paper and will be further studied in the future.

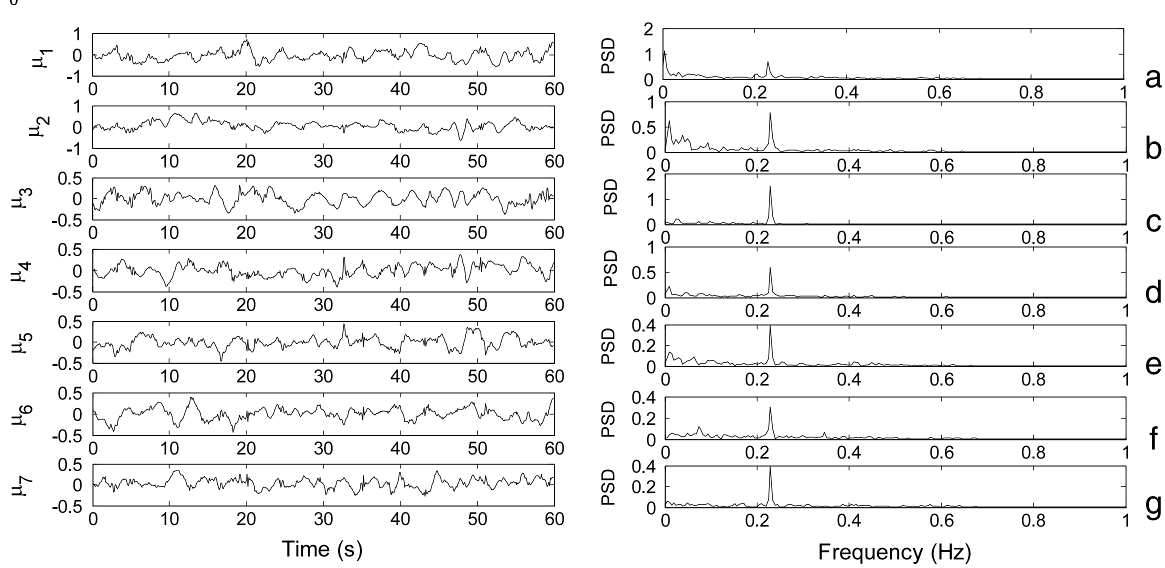

Fig. 21. The first seven principal components and corresponding power spectra: (a) principal component 1, (b) principal component 2, (c) principal component 3, (d) principal component 4, (e) principal component 5, (f) principal component 6, and (g) principal component 7.

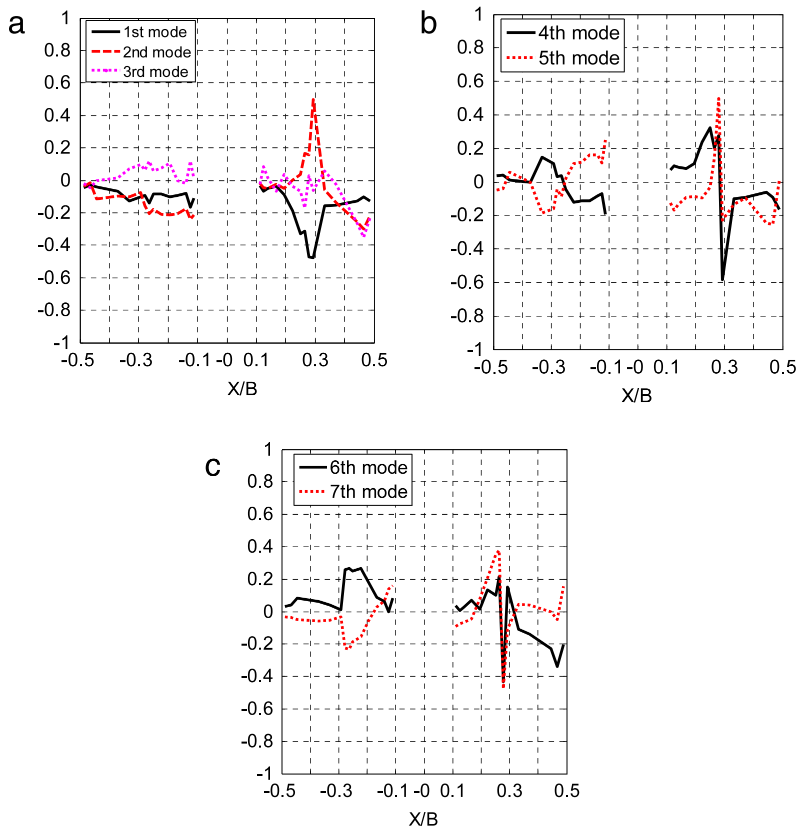

Fig. 23. The first seven mode shapes: (a) 1st, 2nd and 3rd modes, (b) 4th and 5th modes, (c) 6th and 7th modes.

7. POD analysis of fluctuant pressure

The proper orthogonal decomposition (POD) is a powerful tool to analyse the surface pressure of an elastic bluff body and understand the interaction mechanism between flow and bluff body [12–15]. The fluctuant pressure field can be expressed as [12]:

$$C_p(t) = \sum_{k=1}^{n} \mu_k(t)\Phi(k), \tag{3}$$

where n is the number of measured pressure point, $\mu_k(t)$ is the kth order principal component, $\Phi(k)$ is the kth order eigenmode, and $C_p(t)$ is the pressure coefficient of measured point.

The analysed pressures is obtained in the lock-in region of VIV event 3#, and the corresponding wind conditions are as follows: $U = 8.41$ m/s, $\theta = 94.0°$ and $I_u = 4.91\%$. According to the measured pressures, the energy percentage of various eigenmode is calculated by:

$$E_k = \frac{\lambda_k}{\sum_{j=1}^{n} \lambda_j}, \tag{4}$$

where λ_k is the kth order eigenvalue, E_k is the energy percentage of kth order eigenmode.

The contribution of various eigenmode is shown in Fig. 19. It can be seen from Fig. 19 that the first three modes contribute with 41% energy to the pressure field, the next four modes contribute with 23.6% energy, and the total of the first seven modes contribute 64.6%. Fig. 20 presents the integral force of lower surface and the reconstructed signal by using the first seven modes. It can be seen that the reconstructed signal fits well with the original signal only except few points. It indicates that the first seven modes can represent the information of the wind pressure of lower surface. The first seven principal components and corresponding power spectra are shown in Fig. 21. It is observed that all the seven principal components have the same dominant frequency $f = 0.2295$ Hz, which is the same with the frequency of the bridge deck. However, as shown in Fig. 22, the phase lags between displacement of the deck and principal components are different. The first three principal components are nearly in phase with the vertical displacement of deck, while the fourth and sixth principal components are nearly in phase with the velocity of deck and the phase lag of the deck displacement with the fifth and seventh principle components is $-120°$. Fig. 23 shows the first seven mode shapes. It is worth noting that the 1st mode shape is very similar to the mean pressure distribution of lower surface.

8. Spatial–temporal evolution of flow around the bridge deck

The spatial–temporal evolution of the flow around the bridge deck can aid to understanding the whole development process of the vortex-induced oscillation. The dynamic characteristics of the wind flowing around the deck surface can be obtained by analysing the fluctuant pressure measured by the wind pressure sensors. For this purpose, the S transform is employed as the time–frequency analysis tool in this study [16–18]. The S-transform can be described as follows:

$$S(\tau, f) = \int_{-\infty}^{\infty} y(t)\frac{|f|}{\sqrt{2\pi}} e^{-\frac{(\tau-t)^2 f^2}{2}} e^{-i2\pi f t}\,dt, \tag{5}$$

where $y(t)$ is the analysed signal, and f is the frequency.

The locations of analysed pressure points are shown in Fig. 4(c). The beginning stage and lock-in stage of VIV event 3# are selected to analyse the process of spatial–temporal evolution of flow when

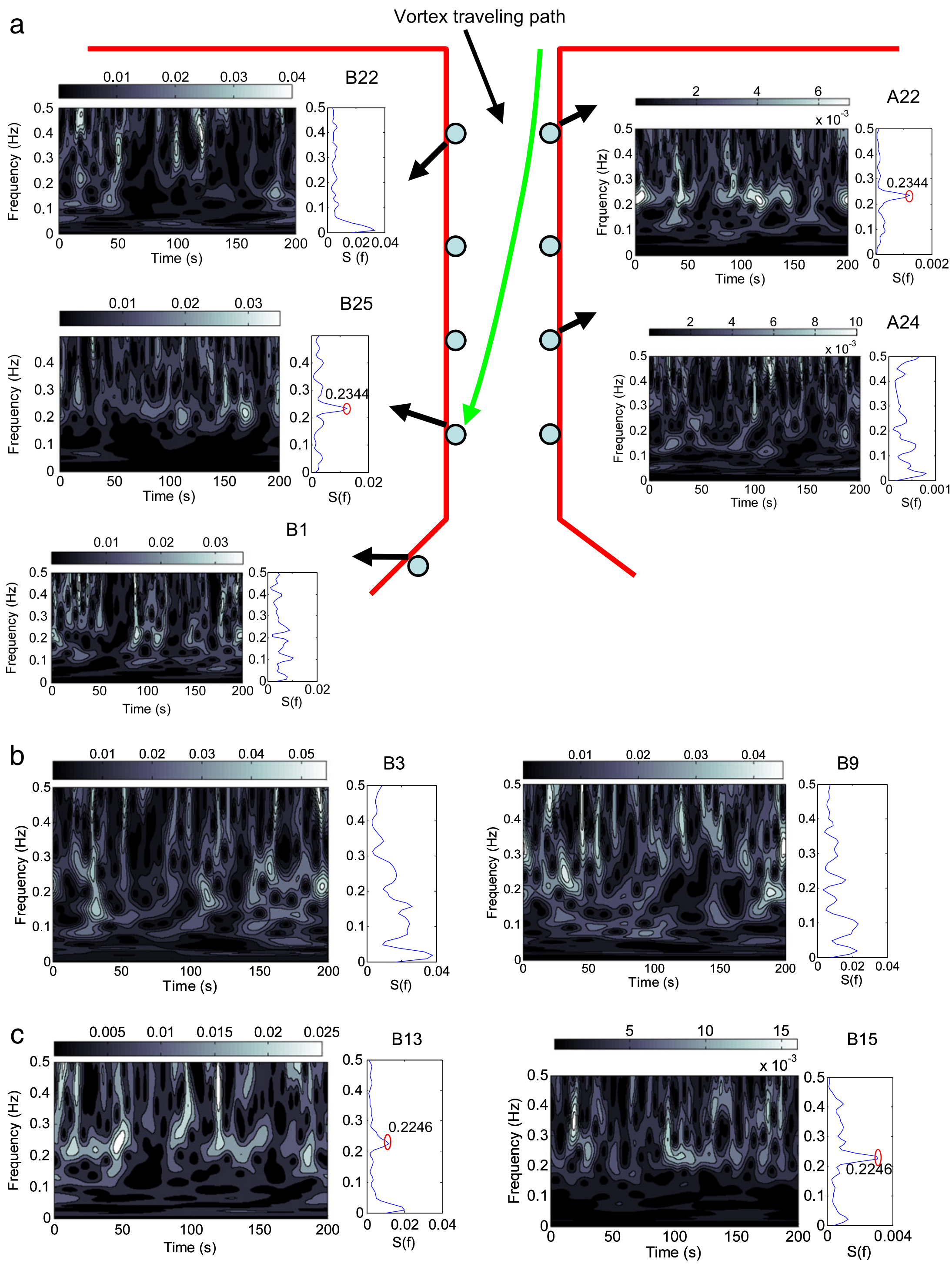

Fig. 24. Time–frequency maps of pressures in the beginning stage of VIV event 3#: (a) at A22, A24, B22, B25, and B1, (b) at B3 and B9, and (c) at B13 and B15.

vortex-induced vibration occurring. Over the beginning stage of VIV event 3# (9:00–9:40 pm, October 29th, 2009), the wind conditions are as follows: $U = 8.47$–8.82 m/s, $\theta = 98.5$–100° and $I_u = 2.25$–5.17%, the response is only dominated by only one single mode with the response frequency of 0.2295 Hz, and the amplitude of oscillation is small. Over the lock-in stage of VIV event 3# (10:25–11:05 pm, October 29th, 2009), the wind conditions are as follows: $U = 8.16$–8.63 m/s, $\theta = 94.0^\circ$–95.5° and $I_u = 3.46\%$–5.80%, due to the strong interaction between the vortex and oscillation, the vortex shedding of flow is controlled by the dramatic motion of the deck, and the response presents the limit-cycle behaviour.

Fig. 24(a) shows the frequency distribution characteristics of the pressures in the gap between the two-box girder at the beginning stage of VIV even 3#. At the locations of A22 and B25, the pressure has a dominant frequency of 0.2344 Hz, while at the locations of A24 and B22, the pressure energy distributions in the time–frequency maps are irregular, and no dominant frequency appears. This phenomenon indicates that regular vortex shedding occurs in the upper sidewall of upstream bridge deck in the

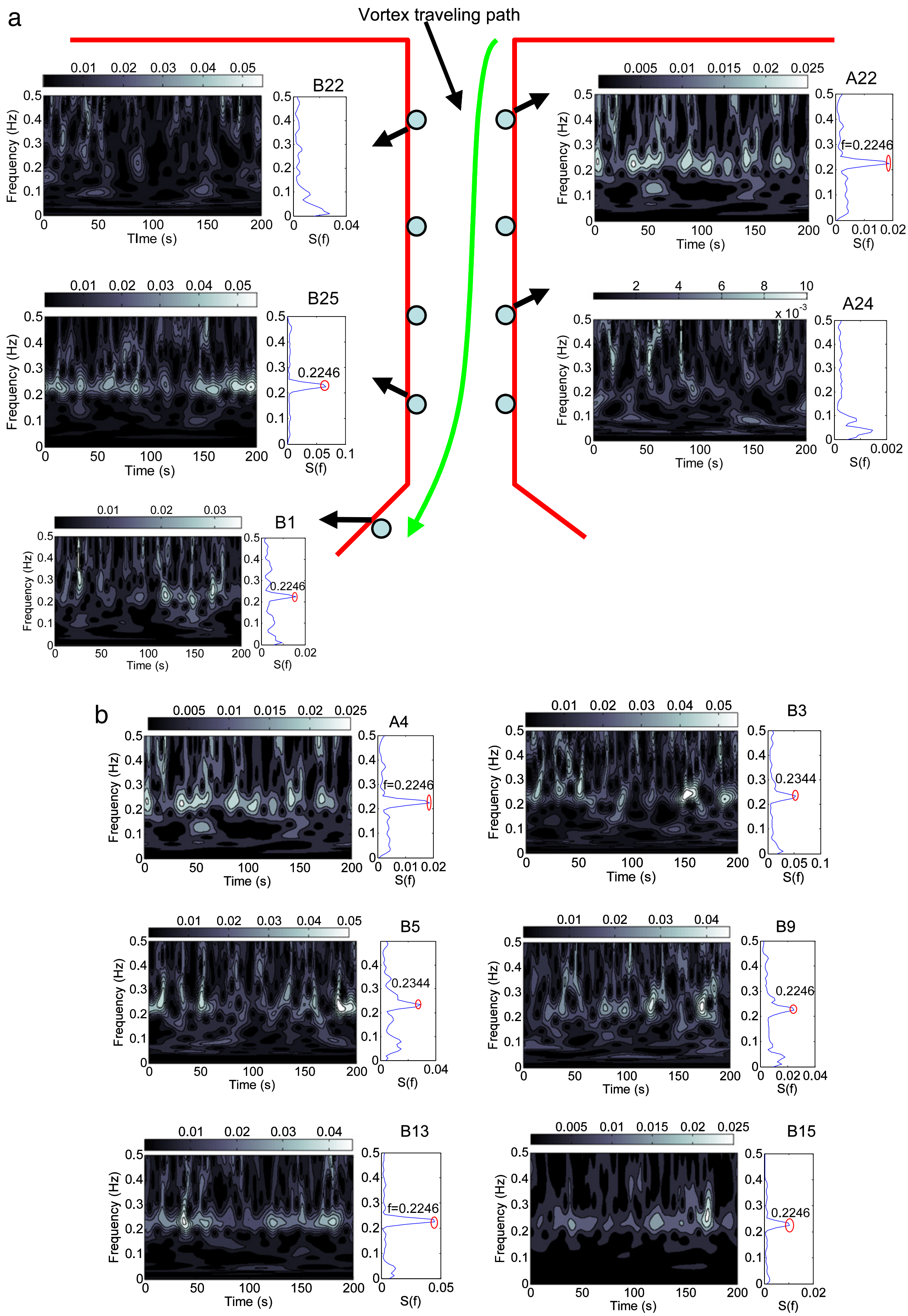

Fig. 25. Time–frequency maps of pressures in the lock-in stage of VIV event 3#: (a)at A22, A24, B22, B25 and B1, and (b) at A4, B3, B5, B9, B13 and B15.

gap, and then the vortex travels to the lower sidewall of the downstream bridge deck (the travel trace is marked in Fig. 24(a)). However, when travelling to the lower sidewall of downstream bridge deck, the strength of vortex is weakened, as observed from the pressure time–frequency maps of A22, B25 and B1. At A22, the eddy with the frequency of 0.2344 Hz appears frequently in the time domain, while at B25, the eddy only appears around 150 s; finally, the eddy that comes from the upper sidewall of

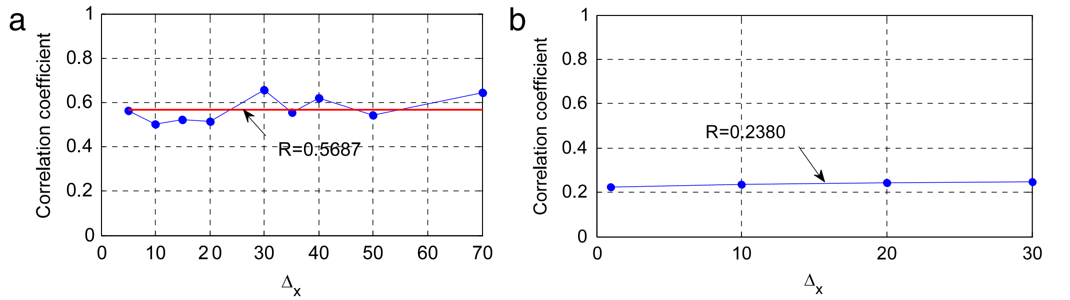

Fig. 26. Span-wise correlation coefficient: (a) at the tailing region of the upper surface, and (b) on the lower surface downstream.

the upstream bridge vanishes at B1. In the following range of B3–B9, there is no dominant frequency, and the pressures are broadband processes, as shown in Fig. 24(b). However, in the trailing region of the lower surface, the pressures again have one dominant frequency of 0.2246 Hz, as shown in Fig. 24(c), indicating that regular vortex shedding phenomena appears in the trailing region of downstream bridge deck once again.

In the lock-in stage of VIV event 3#, due to the strong interaction between the vortex and oscillation, the vortex shedding regions extend, and the strength of vortex is increasing. In the gap of the deck, the time–frequency maps of A22 and B25 as shown in Fig. 25(a) indicate that the flow behaves distinct vortex shedding phenomena, and the eddy with a frequency of 0.2246 Hz appears continuously in the time domain; in particular for the pressure at A22, the energy of the vortex is uniformly distributed over the entire time domain. At B1, the pressure is still dominated by the resonance frequency. Unlike in the beginning stage of the vortex-induced vibration event 3#, the vortex does not vanish at this location. At position A4, which locates in the tailing of the upstream deck, regular vortex shedding occurs, as presented in the time–frequency maps of the pressures of Fig. 25(b), and the corresponding frequency is 0.2246 Hz, which is close to the resonance response frequency of 0.2297 Hz. In the following ranges of B3, B5, B9, B13 and B15, the dominant frequency of 0.2246 Hz is observed in the time–frequency maps, as shown in Fig. 25(b). It indicates that the vortex shedding phenomenon extends to the entire lower surface of downstream deck.

The above analysis indicates that in the beginning stage of vortex-induced vibration event 3#, regular vortex shedding only occurs in the gap and at the tailing region of the downstream bridge deck. However, in the lock-in stage, due to the large vibration of the deck, the vortex shedding phenomena extends from the gap to the entire lower surface of downstream deck and the tail of upstream deck, the vortex shedding regions in the gap and lower surface link together and the VIV is much stronger.

9. The span-wise correlation of wind pressure in the lock-in range

The span-wise correlation of the wind pressure is an important issue for long-span bridges. In this wind monitoring system, the wind pressure sensors are installed in different sections along the longitudinal bridge axis, as shown in Fig. 4(b). The analysed case is selected from the lock-in range of the VIV event 3#, the duration of analysed data is 10 min (10:40–10:50 pm, October 29th, 2009), the resonance frequency is 0.2295 Hz, and the corresponding wind conditions are as follows: $U = 8.41$ m/s, $\theta = 94.0°$ and $I_u = 4.91\%$.

The correlation of wind pressure is defined as

$$R(x_1, x_2) = \frac{\mathrm{COV}(p(x_1, t), p(x_2, t))}{\sqrt{\sigma(p(x_1, t))}\sqrt{\sigma(p(x_1, t))}}, \qquad (6)$$

where $\mathrm{COV}(p(x_1, t), p(x_2, t))$ represents the covariance of pressure signals $p(x_1, t)$ and $p(x_2, t)$, and $\sigma(p(x_1, t))$ and $\sigma(p(x_2, t))$ represent the variances of $p(x_1, t)$ and $p(x_2, t)$, respectively.

The wind pressures at the tailing region of the upper surface and on the lower surface of downstream deck are selected to investigate the span-wise correlation. The span-wise correlations at the two locations of the box girder are calculated and shown in Fig. 26. The values in these figures indicate that the correlations of the pressure almost do not decrease with the increase in distance, which due to that the dramatic vibrations control the vortex shedding. At the upper tailing region of downstream deck, the value of correlation is 0.5687; on the lower surface of downstream deck, the value of correlation is 0.2380.

The correlation in the frequency domain is also important, in particular for the vortex-induced oscillation occurring. In this paper, the coherent function is adopted to quantify the span-wise correlation of the wind pressure in the frequency space. The coherent function is defined as follows:

$$C(x_1, x_2, f) = \frac{|S(p(x_1, t), p(x_2, t))|^2}{S(p(x_1, t), p(x_1, t))S(p(x_2, t), p(x_2, t))}, \qquad (7)$$

where $C(x_1, x_2, f)$ is the coherence function, $S(p(x_1, t), p(x_2, t))$ is the crossing power density spectra of $p(x_1, t)$ and $p(x_2, t)$, and $S(p(x_1, t), p(x_1, t))$ and $S(p(x_2, t), p(x_2, t))$ are auto-power density spectra.

The calculation results are shown in Fig. 27(a) and (b). The wind pressure has a high coherence at a resonance frequency of 0.2246 Hz. It indicates that the pressure distributions along the longitudinal bridge axis are controlled by the vibration.

10. Conclusions

In this paper, the vortex-induced vibration of a suspension bridge is investigated based on field measurements. The following conclusions can be obtained from this study.

It is observed that vertical vortex-induced resonance of the deck might occur with the wind direction nearly perpendicular to the longitudinal bridge axis and a low turbulence intensity in the wind speed range of 6–11 m/s. Analysis of the process of the VIV events 3# indicates that the turbulence intensity is a key factor impacting the oscillation amplitude and lock-in region. If the turbulence intensity remains at a low value for a long time (in this paper, the turbulence is below 2%), large amplitude oscillations can occur.

The pressure sensors, which are installed around the quarter span section, provide the full-scale mean pressure distribution over the lock-in range. The result indicates that all the pressure coefficients on the lower surface of upstream bridge deck are negative, and on the corner, the pressure reaches the max negative value, then the pressure recovers rapidly in the following range. On the lower surface of downstream bridge deck, the pressure coefficients are close to zero. In the gap of the bridge deck, the mean pressure coefficients are also close to zero.

The spatial–temporal evolution of flow on the lower surface of the bridge deck is obtained through the time–frequency maps of

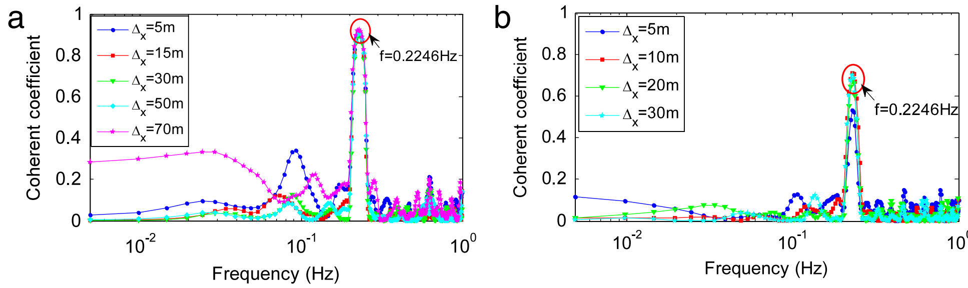

Fig. 27. The coherent coefficients of the wind pressure along the longitudinal bridge axis: (a) at the trailing edge of upper surface, and (b) on the lower surface downstream.

the pressure on the deck surface. In the beginning stage of VIV, the vortex shedding phenomena only occur in the gap of two bridge deck and at the tail of downstream deck. However, in the lock-in stage, the vortex shedding phenomena extends to the entire lower surface of downstream deck and the tail of upstream deck, and the vortex shedding regions in the gap and lower surface link together. The correlations of the pressure along the bridge axis are very high and do not decrease with the increase in distance, which may be attributed to the findings that the correlations of the pressure are controlled by the large vortex-induced vibration.

Acknowledgements

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